

The Definition of the Information Matrix for the UWB Factor

The information matrix for the UWB factor in the manuscript is illustrated in the figure below. For clarity, the short sliding window states $\mathbf{x}$ consist of 3 keyframes(x_0, x_1, x_2), the visual reprojections λ consist of 4 feature points ($\lambda_0, \lambda_1, \lambda_2, \lambda_3$), and the long sliding window states $\mathbf{w}$ consist of 6 key trajectory points ($w_0, w_1, w_2, w_3, w_4, w_5$).

	x_0	x_1	x_2	λ_0	λ_1	λ_2	λ_3	w_0	w_1	w_2	w_3	w_4	w_5
x_0	Dark Green	Light Green	White	Blue	White	Blue	White	White	White	White	Gray	Gray	Gray
x_1	Light Green	Dark Green	Light Green	Blue	Blue	White	White	White	White	White	White	White	White
x_2	White	Light Green	Dark Green	White	Blue	Blue	Blue	White	White	White	White	White	White
λ_0	Blue	Blue	White	Yellow	White	White	White	White	White	White	White	White	White
λ_1	White	Blue	Blue	White	Yellow	White	White	White	White	White	White	White	White
λ_2	Blue	White	Blue	White	White	Yellow	White	White	White	White	White	White	White
λ_3	White	White	Blue	White	White	White	Yellow	White	White	White	White	White	White
w_0	White	White	White	White	White	White	White	Orange	Orange	Orange	Orange	White	White
w_1	White	White	White	White	White	White	White	Orange	Orange	Orange	Orange	Orange	White
w_2	White	White	White	White	White	White	White	Orange	Orange	Orange	Orange	Orange	Orange
w_3	Gray	White	White	White	White	White	White	Orange	Orange	Orange	Orange	Orange	Orange
w_4	Gray	White	White	White	White	White	White	White	Orange	Orange	Orange	Orange	Orange
w_5	Gray	White	White	White	White	White	White	White	White	Orange	Orange	Orange	Orange

As shown in the figure, this information matrix has a sparsity pattern similar to that of conventional VIO. On this basis, the UWB factor (Eq. (5)) introduces additional structural variations, mainly reflected in the coupling between the long and cross sliding windows, as follows:

- 1) The long-window ranging factor R_l acts on key trajectory points at the same time, forming diagonal constraints in the (w, w) sub-block (orange part); combined with pose graph (PG) constraints, it also forms a banded sparse structure near the main diagonal.
- 2) The short-window ranging factor R_s only affects the short sliding window states and primarily enhances the diagonal information of the position components (**dark green** part in the (x, x) sub-block), which does not alter the overall sparsity pattern.
- 3) Cross-window coupling is only introduced by pose graph constraints, corresponding to the (x, w) sub-block (gray part), with non-zero entries only between the oldest short-window keyframe x_0 and the most recent long-window key trajectory points (e.g., w_3, w_4, w_5).
- 4) Since the UWB residuals only relate to positional information and do not involve visual feature parameters, the remaining (λ, w) sub-blocks are all zero.